

LV Cable Sizing Calculation

Project	Sample Plant (invented data)
Date	2026-09-22
Code	IEC 60364-5-52:2009 (values as reproduced in IS 732:2019)
Tool version	cablecalc 0.1.0
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Checked by: _____

Symbols

I _b	Design current: the current the load draws (A).
I _n	Rated current of the overload protective device (A); none if no device was entered.
I _t	Tabulated current-carrying capacity: the rating read from the code's table for this size, conductor, insulation, installation method and number of loaded conductors, under the table's reference conditions of 30 deg C air or 20 deg C ground, soil thermal resistivity 2.5 K.m/W and one circuit on its own (A).
C _a	Correction factor for the air temperature, or for the ground temperature for buried cables.
C _s	Correction factor for the soil thermal resistivity (buried cables only).
C _g	Group reduction factor for the other circuits run alongside.
I _z	Current-carrying capacity in the actual installation: It multiplied by the correction factors (A). It must be at least I _b , and at least I _n where a device is entered.
dU	Voltage drop along the cable (V), also given as a percentage of the supply voltage.
b	Circuit factor: $\sqrt{3}$ for three-phase (on the line voltage), 2 for single-phase.
L	Route length, one way (m).
R, X	Resistance and reactance of the conductor per kilometre (ohm/km).
cos phi, sin phi	cos phi is the power factor of the load; $\sin \phi = \sqrt{1 - \cos^2 \phi}$.
S	Cross-sectional area of the conductor (mm ²).
I	Prospective short-circuit current at the cable (kA).
t	Time for the protective device to clear that fault (s).
k	Factor for the conductor and insulation, from IS 732 Table 3.

Cable F-01 Pump motor: 3C x 16 mm² Cu PVC

16 mm² passes all checks

Current-carrying capacity

Formula	$I_z = I_t \times C_a \times C_g \geq \max(I_b, I_n)$
Values	$I_b = 36.01 \text{ A}$, $I_n = 50 \text{ A}$, $I_t = 76 \text{ A}$, $C_a = 0.87$, $C_g = 0.85$, $I_z = 56.20 \text{ A}$
Required	50 A
Actual	56.2 A
Result	PASS
Sources	IEC 60364-5-52:2009 Table B.52.4, method C, values as reproduced in IS 732:2019 Table 23 IEC 60364-5-52:2009 Table B.52.14, PVC, air temperature, values as reproduced in IS 732:2019 Table 33 IEC 60364-5-52:2009 Table B.52.17, item 2, single layer on wall, floor or unperforated tray, values as reproduced in IS 732:2019 Table 36

Voltage drop

Formula	$dU = b \times I_b \times L \times (R \cos \phi + X \sin \phi)$; $b = \sqrt{3}$ (3-phase, on line voltage) or 2 (single-phase)
Values	$R = 1.4062 \text{ ohm/km}$, $X = 0.08 \text{ ohm/km}$, $L = 60 \text{ m}$, $dU = 4.63 \text{ V}$
Required	5 %
Actual	1.12 %
Result	PASS
Sources	IEC 60364-5-52:2009 Annex G method, as reproduced in IS 732:2019 Annex Y: $R = \rho_{01} / S$ with $\rho_{01} = 0.0225 \text{ ohm.mm}^2/\text{m}$; $X = 0.08 \text{ mohm/m}$

Short-circuit withstand

Formula	$S \geq I \times \sqrt{t} / k$
Values	$I = 5 \text{ kA}$, $t = 0.1 \text{ s}$, $k = 115$
Required	13.75 mm ²
Actual	16 mm ²
Result	PASS
Sources	IEC 60364-4-43:2008 k values, as reproduced in IS 732:2019 Table 3, conductors up to 300 mm ²

Cable F-02 Yard lighting: 2 x 1C x 6 mm² Cu PVC

6 mm² passes all checks

Current-carrying capacity

Formula	$I_z = I_t \times C_a \times C_g \geq \max(I_b, I_n)$
Values	$I_b = 10.00 \text{ A}$, $I_n = 16 \text{ A}$, $I_t = 41 \text{ A}$, $C_a = 0.94$, $C_g = 0.8$, $I_z = 30.83 \text{ A}$
Required	16 A
Actual	30.83 A
Result	PASS
Sources	IEC 60364-5-52:2009 Table B.52.2, method B1, values as reproduced in IS 732:2019 Table 21 IEC 60364-5-52:2009 Table B.52.14, PVC, air temperature, values as reproduced in IS 732:2019 Table 33 IEC 60364-5-52:2009 Table B.52.17, item 1, bunched in air, on a surface, embedded or enclosed, values as reproduced in IS 732:2019 Table 36

Voltage drop

Formula	$dU = b \times I_b \times L \times (R \cos \phi + X \sin \phi)$; $b = \sqrt{3}$ (3-phase, on line voltage) or 2 (single-phase)
Values	$R = 3.75 \text{ ohm/km}$, $X = 0.08 \text{ ohm/km}$, $L = 95 \text{ m}$, $dU = 6.48 \text{ V}$
Required	3 %
Actual	2.82 %
Result	PASS
Sources	IEC 60364-5-52:2009 Annex G method, as reproduced in IS 732:2019 Annex Y: $R = \rho_{01} / S$ with $\rho_{01} = 0.0225 \text{ ohm.mm}^2/\text{m}$; $X = 0.08 \text{ mohm/m}$

Short-circuit withstand

Formula	$S \geq I \times \sqrt{t} / k$
Values	$I = 1 \text{ kA}$, $t = 0.1 \text{ s}$, $k = 115$
Required	2.75 mm ²
Actual	6 mm ²
Result	PASS
Sources	IEC 60364-4-43:2008 k values, as reproduced in IS 732:2019 Table 3, conductors up to 300 mm ²

Cable F-03 Board tail: 4C x 120 mm² Cu XLPE

120 mm² passes all checks

Current-carrying capacity

Formula	$I_z = I_t \times C_a \times C_g \geq \max(I_b, I_n)$
Values	$I_b = 40.00 \text{ A}$, $I_n = 50 \text{ A}$, $I_t = 346 \text{ A}$, $C_a = 1$, $C_g = 1$, $I_z = 346.00 \text{ A}$
Required	50 A
Actual	346 A
Result	PASS
Sources	IEC 60364-5-52:2009 Table B.52.12, method E, multi-core, 3 loaded conductors, values as reproduced in IS 732:2019 Table 31 IEC 60364-5-52:2009 Table B.52.14, XLPE, air temperature, values as reproduced in IS 732:2019 Table 33 IEC 60364-5-52:2009 Table B.52.17, item 4, single layer on perforated tray, values as reproduced in IS 732:2019 Table 36

Voltage drop

Formula	$dU = b \times I_b \times L \times (R \cos \phi + X \sin \phi)$; $b = \sqrt{3}$ (3-phase, on line voltage) or 2 (single-phase)
Values	$R = 0.1875 \text{ ohm/km}$, $X = 0.08 \text{ ohm/km}$, $L = 10 \text{ m}$, $dU = 0.14 \text{ V}$
Required	5 %
Actual	0.03 %
Result	PASS
Sources	IEC 60364-5-52:2009 Annex G method, as reproduced in IS 732:2019 Annex Y: $R = \rho_{01} / S$ with $\rho_{01} = 0.0225 \text{ ohm.mm}^2/\text{m}$; $X = 0.08 \text{ mohm/m}$

Short-circuit withstand

Formula	$S \geq I \times \sqrt{t} / k$
Values	$I = 25 \text{ kA}$, $t = 0.4 \text{ s}$, $k = 143$
Required	110.57 mm ²
Actual	120 mm ²
Result	PASS
Sources	IEC 60364-4-43:2008 k values, as reproduced in IS 732:2019 Table 3, conductors up to 300 mm ²

Cable F-04 Store building: 3.5C x 185 mm² Al XLPE

185 mm² passes all checks

Current-carrying capacity

Formula	$I_z = I_t \times C_a \times C_s \times C_g \geq \max(I_b, I_n)$
Values	$I_b = 150.00 \text{ A}$, $I_n = 160 \text{ A}$, $I_t = 250 \text{ A}$, $C_a = 0.93$ (ground), $C_s = 1$, $C_g = 0.75$, $I_z = 174.38 \text{ A}$
Required	160 A
Actual	174.38 A
Result	PASS
Sources	IEC 60364-5-52:2009 Table B.52.5, method D2, values as reproduced in IS 732:2019 Table 24 IEC 60364-5-52:2009 Table B.52.15, XLPE, ground temperature, values as reproduced in IS 732:2019 Table 34 IEC 60364-5-52:2009 Table B.52.16, direct buried cables, values as reproduced in IS 732:2019 Table 35; IEC table number by the consistent IS-to-IEC numbering IEC 60364-5-52:2009 Table B.52.18, cables laid direct in the ground, nil clearance column, values as reproduced in IS 732:2019 Table 37 (cables or ducts touching assumed)

Voltage drop

Formula	$dU = b \times I_b \times L \times (R \cos \phi + X \sin \phi)$; $b = \sqrt{3}$ (3-phase, on line voltage) or 2 (single-phase)
Values	$R = 0.19459 \text{ ohm/km}$, $X = 0.08 \text{ ohm/km}$, $L = 150 \text{ m}$, $dU = 8.09 \text{ V}$
Required	5 %
Actual	1.95 %
Result	PASS
Sources	IEC 60364-5-52:2009 Annex G method, as reproduced in IS 732:2019 Annex Y: $R = \rho_{01} / S$ with $\rho_{01} = 0.036 \text{ ohm.mm}^2/\text{m}$; $X = 0.08 \text{ mohm/m}$

Short-circuit withstand

Formula	$S \geq I \times \sqrt{t} / k$
Values	$I = 10 \text{ kA}$, $t = 0.3 \text{ s}$, $k = 94$
Required	58.27 mm ²
Actual	185 mm ²
Result	PASS
Sources	IEC 60364-4-43:2008 k values, as reproduced in IS 732:2019 Table 3, conductors up to 300 mm ²

Cable F-05 Main incomer: 4 x 1C Cu PVC, no size passes

No size passes. Results shown are for the largest size in the table (300 mm²).

Current-carrying capacity

Formula	$I_z = I_t \times C_a \times C_g \geq \max(I_b, I_n)$
Values	$I_b = 600.00 \text{ A}$, $I_n = 630 \text{ A}$, $I_t = 464 \text{ A}$, $C_a = 0.87$, $C_g = 0.79$, $I_z = 318.91 \text{ A}$
Required	630 A
Actual	318.91 A
Result	FAIL
Sources	IEC 60364-5-52:2009 Table B.52.4, method C, values as reproduced in IS 732:2019 Table 23 IEC 60364-5-52:2009 Table B.52.14, PVC, air temperature, values as reproduced in IS 732:2019 Table 33 IEC 60364-5-52:2009 Table B.52.17, item 2, single layer on wall, floor or unperforated tray, values as reproduced in IS 732:2019 Table 36

Voltage drop

Formula	$dU = b \times I_b \times L \times (R \cos \phi + X \sin \phi)$; $b = \sqrt{3}$ (3-phase, on line voltage) or 2 (single-phase)
Values	$R = 0.075 \text{ ohm/km}$, $X = 0.08 \text{ ohm/km}$, $L = 40 \text{ m}$, $dU = 4.26 \text{ V}$
Required	5 %
Actual	1.03 %
Result	PASS
Sources	IEC 60364-5-52:2009 Annex G method, as reproduced in IS 732:2019 Annex Y: $R = \rho_{01} / S$ with $\rho_{01} = 0.0225 \text{ ohm.mm}^2/\text{m}$; $X = 0.08 \text{ mohm/m}$

Short-circuit withstand

Formula	$S \geq I \times \sqrt{t} / k$
Values	$I = 25 \text{ kA}$, $t = 0.4 \text{ s}$, $k = 115$
Required	137.49 mm ²
Actual	300 mm ²
Result	PASS
Sources	IEC 60364-4-43:2008 k values, as reproduced in IS 732:2019 Table 3, conductors up to 300 mm ²

Summary

Cable	Construction	Size (mm ²)	Governing check	Result
F-01 Pump motor	3C Cu PVC	16	Short-circuit withstand	PASS
F-02 Yard lighting	2 x 1C Cu PVC	6	Voltage drop	PASS
F-03 Board tail	4C Cu XLPE	120	Short-circuit withstand	PASS
F-04 Store building	3.5C Al XLPE	185	Current-carrying capacity	PASS
F-05 Main incomer	4 x 1C Cu PVC	-	-	FAIL

Governing check: the check that the next smaller size fails (the one failing by the widest margin, if more than one does). A dash means the smallest size in the table already passes.

Limitations

Low-voltage cables up to 1.1 kV only.

Only the installation methods and sizes present in the shipped tables are supported. An ambient temperature, number of grouped circuits or soil thermal resistivity between tabulated rows takes the next higher row, as stated in the Sources line; values are never interpolated.

For buried cables (methods D1 and D2) the ambient temperature is the ground temperature, and grouping assumes the cables or ducts are touching.

Group reduction factors assume a group of similar, equally loaded cables. Groups containing different sizes are not covered; for those, IS 732 clause S-5.1 gives $F = 1/\sqrt{n}$.

Overload protection: condition (1) of IS 732 clause 4.4.4.1, $I_b \leq I_n \leq I_z$, is checked. Condition (2), $I_2 \leq 1.45 \times I_z$, is not calculated. It is met by a device whose conventional tripping current I_2 is not more than $1.45 \times I_n$, such as a circuit-breaker to IEC 60898-1; for any other device the engineer must check condition (2).

Short-circuit withstand is checked as $S \geq I \times \sqrt{t} / k$ (IS 732 clause 4.4.5.5.2, equation 3). The same clause requires $k^2 \times S^2$ to exceed the let-through energy $I^2 t$ quoted by the device manufacturer where the device operates in less than 0.1 s or is current-limiting. That check is not calculated; the engineer must make it from the manufacturer's data.

The fault current and fault time entered must be the worst pair for a fault anywhere along the cable (IS 732 clause 4.4.5.5.2). A fault at the far end is smaller but may take longer to clear, so it can be the more onerous case. The tool checks only the pair entered.

Each cable is one run per phase. Cables in parallel are not covered, and neither are single-core cables in free air (methods F and G).

For a 3.5C cable the size given is that of the phase conductors; the reduced neutral follows the cable standard.

Harmonic derating is not calculated.

This is a sample tool, not a design service. Results must be checked by a qualified engineer.